

scope of this book.) For example, if `value` stored the `date` object returned by `datetime.date(2021, 1, 1)`, then `value.__class__.__qualname__` would be the string `'date'`. The exception messages use `value.__class__.__qualname__` ⑦ to get a string of the value object's name. The class name makes the error message more useful to the programmer reading it, because it identifies not only that the `value` argument was not the right type, but what type it was and what type it should be.

You'll need to copy the code for the getter and setter for `_galleons` to use for the `_sickles` and `_knuts` attributes as well. Their code is identical except they use the `_sickles` and `_knuts` attributes, instead of `_galleons`, as backing variables.

Read-Only Properties

Your objects might need some read-only properties that can't be set with the assignment operator `=`. You can make a property read-only by omitting the setter and deleter methods.

For example, the `total()` method in the `WizCoin` class returns the value of the object in knuts. We could change this from a regular method to a read-only property, because there is no reasonable way to set the `total` of a `WizCoin` object. After all, if you set `total` to the integer `1000`, does this mean 1,000 knuts? Or does it mean 1 galleon and 493 knuts? Or does it mean some other combination? For this reason, we'll make `total` a read-only property by adding the code in bold to the `wizcoin.py` file:

```
@property
def total(self):
    """Total value (in knuts) of all the coins in this WizCoin object."""
    return (self.galleons * 17 * 29) + (self.sickles * 29) + (self.knuts)
```

Note that there is no setter or deleter method for `total`.

After you add the `@property` function decorator in front of `total()`, Python will call the `total()` method whenever `total` is accessed. Because there is no setter or deleter method, Python raises `AttributeError` if any code attempts to modify or delete `total` by using it in an assignment or `del` statement, respectively. Notice that the value of the `total` property depends on the value